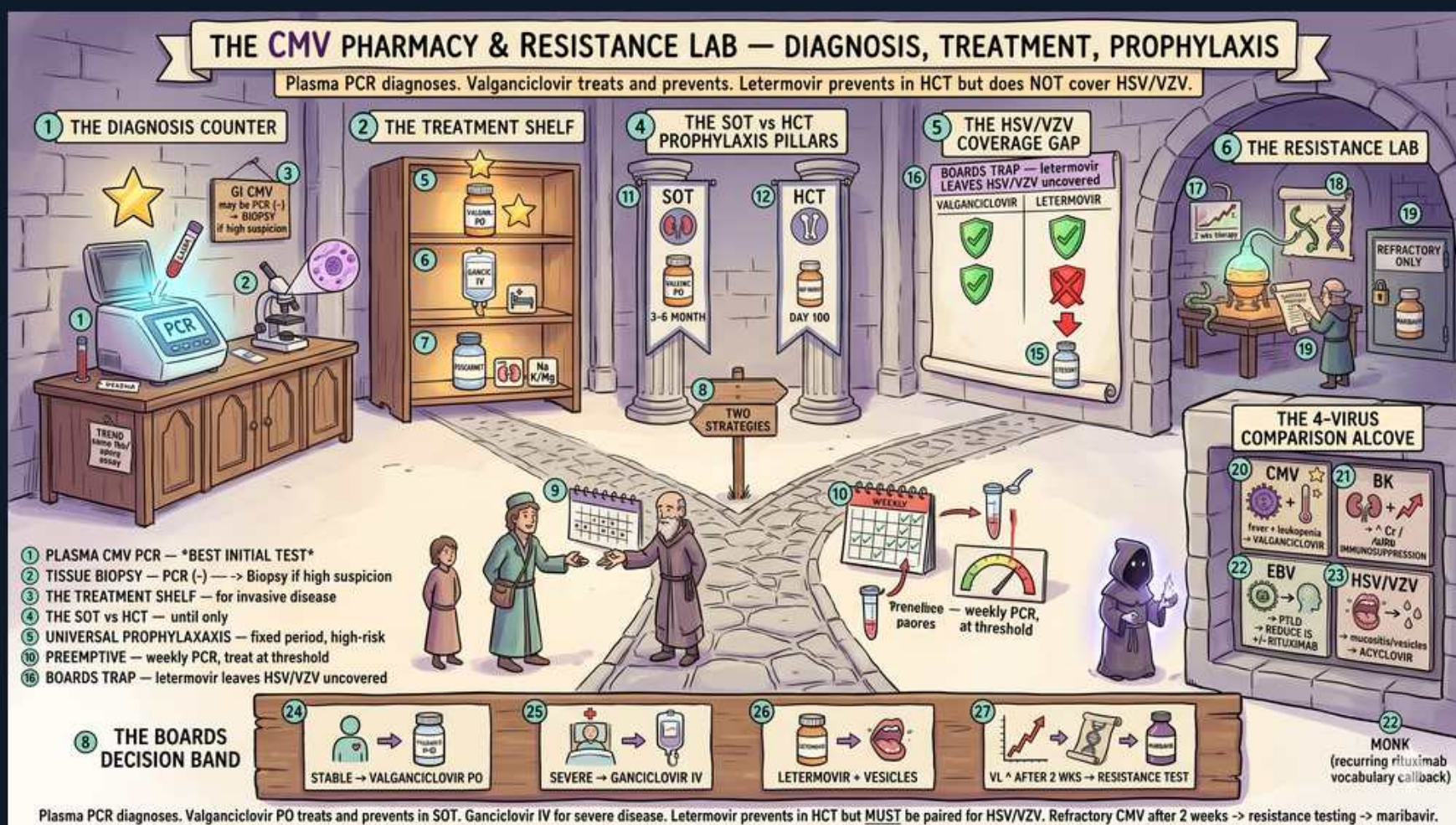


CMV After Kidney Transplant — Diagnosis, Treatment & Prophylaxis



1. Plasma CMV PCR — best initial test

WHAT YOU SEE

Diagnosis counter with PCR machine and quantitative readout

WHAT IT ENCODES

Quantitative plasma CMV DNA PCR (viral load) is the FIRST and best test for suspected CMV — both for diagnosis and to monitor response to therapy. Rising or persistently high viral load signals treatment failure or resistance.

BOARD TRAP

CMV IgM/IgG serology is the wrong answer in transplant — antibody is unreliable under immunosuppression; PCR viral load is the standard.

2. Tissue biopsy — invasive/tissue-invasive disease

WHAT YOU SEE

Biopsy note when PCR is low/negative but disease suspected

WHAT IT ENCODES

When tissue-invasive disease (especially GI CMV colitis) is suspected but plasma PCR is low or negative, biopsy with histology (owl-eye inclusions/immunostain) confirms. GI disease can be PCR-negative in blood.

BOARD TRAP

A negative blood PCR does NOT exclude tissue-invasive GI CMV — biopsy is required, not reassurance.

3. Valganciclovir — oral first-line treatment

WHAT YOU SEE

Treatment shelf with valganciclovir bottle

WHAT IT ENCODES

Oral valganciclovir is first-line for non-severe CMV disease and is the prodrug of ganciclovir. It both treats active disease and is used for prophylaxis. Continue until viral load clears (often two consecutive negative PCRs).

BOARD TRAP

Valganciclovir is for mild/moderate disease and outpatient management — severe or GI disease needs IV ganciclovir, not oral val.

4. IV ganciclovir — severe disease

WHAT YOU SEE

Boards decision band: severe → ganciclovir IV

WHAT IT ENCODES

Severe, life-threatening, or poorly-absorbed (GI) CMV disease is treated with IV ganciclovir. Step down to oral valganciclovir once improving and absorbing.

BOARD TRAP

Severe pneumonitis or colitis treated with oral valganciclovir risks underdosing — use IV ganciclovir first.

5. Universal prophylaxis — D+/R- highest risk

WHAT YOU SEE

Universal prophylaxis arrow, weekly PCR for preemptive

WHAT IT ENCODES

Donor-positive/recipient-negative (D+/R-) is the HIGHEST-risk CMV mismatch and warrants universal valganciclovir prophylaxis. Alternative strategy is preemptive: weekly PCR surveillance, treat when threshold crossed.

BOARD TRAP

D-/R- is the LOWEST risk (no prophylaxis needed); the dangerous mismatch is D+/R-, not the reverse.

6. Preemptive strategy — weekly PCR, treat at threshold

WHAT YOU SEE

Threshold gauge with weekly PCR scores

WHAT IT ENCODES

Preemptive therapy monitors viral load weekly and starts antivirals only when a viral-load threshold is reached, sparing drug toxicity. Universal prophylaxis instead gives drug to all at-risk for a fixed period.

BOARD TRAP

Preemptive ≠ prophylaxis — preemptive treats viremia early on surveillance; it does not prevent reactivation the way universal prophylaxis does.

7. Letermovir — prevents but does NOT treat

WHAT YOU SEE

Coverage-gap pillar: letermovir leaves HSV/VZV uncovered

WHAT IT ENCODES

Letermovir (terminase inhibitor) is used for CMV PROPHYLAXIS, mainly in HCT/stem-cell recipients. It does not treat established CMV disease and has NO activity against HSV or VZV.

BOARD TRAP

Board trap — letermovir 'covers' CMV only; it leaves HSV/VZV uncovered and is not a treatment drug. Valganciclovir covers HSV/VZV too.

8. Ganciclovir/valganciclovir cover HSV & VZV

WHAT YOU SEE

Coverage chart linking val-/ganciclovir to herpesviruses

WHAT IT ENCODES

Unlike letermovir, ganciclovir and valganciclovir also suppress HSV and VZV, so patients on val-prophylaxis don't need separate acyclovir. This dual coverage is a frequent exam discriminator.

BOARD TRAP

Don't add acyclovir for a patient already on valganciclovir prophylaxis — it's redundant; letermovir patients DO still need HSV/VZV coverage.

9. Ganciclovir resistance — UL97/UL54 mutations

WHAT YOU SEE

Resistance lab with mutation panel

WHAT IT ENCODES

Failure to clear viremia on adequate ganciclovir suggests resistance, classically UL97 (and UL54) mutations. Send genotypic resistance testing. Options then include foscarnet, cidofovir, or maribavir.

BOARD TRAP

Persistently rising PCR despite therapy is resistance, not under-dosing alone — test for UL97 before assuming non-adherence.

10. Foscarnet / cidofovir — resistant CMV salvage

WHAT YOU SEE

Resistance lab salvage agents (foscarnet, cidofovir)

WHAT IT ENCODES

For ganciclovir-resistant CMV, foscarnet (nephrotoxic, electrolyte wasting) or cidofovir (nephrotoxic) are salvage; maribavir is a newer oral option for refractory/resistant disease.

BOARD TRAP

Foscarnet's classic toxicity is electrolyte derangement (low Ca/Mg/K) and renal injury — not marrow suppression (that's ganciclovir).

11. Ganciclovir/valganciclovir myelosuppression

WHAT YOU SEE

Shelf note on leukopenia from val/ganciclovir

WHAT IT ENCODES

Dose-limiting toxicity of ganciclovir/valganciclovir is bone-marrow suppression — neutropenia/leukopenia. Monitor CBC; reduce dose or add G-CSF. This overlaps with MMF-induced cytopenia.

BOARD TRAP

New neutropenia on prophylaxis is often the valganciclovir (or co-administered MMF), not CMV marrow invasion — adjust the drug.

12. BK virus — distinct from CMV (alcove)

WHAT YOU SEE

4-virus comparison alcove: CMV vs BK vs EBV vs HSV/VZV

WHAT IT ENCODES

BK polyomavirus causes nephropathy/graft dysfunction and is managed by REDUCING immunosuppression, not antivirals. It is a key contrast to CMV, which is treated with antivirals.

BOARD TRAP

Don't reach for ganciclovir for BK nephropathy — BK has no good antiviral; the answer is lowering immunosuppression.

13. EBV → PTLD link

WHAT YOU SEE

EBV node in the virus alcove

WHAT IT ENCODES

EBV drives post-transplant lymphoproliferative disorder (PTLD). EBV viral load monitoring matters in high-risk recipients; management is reducing immunosuppression ± rituximab.

BOARD TRAP

EBV is the PTLT driver, not CMV — CMV is associated with allograft injury and indirect immunomodulation, not lymphoma.

14. Boards decision band

WHAT YOU SEE

Bottom band: stable→val PO, severe→IV ganciclovir, HCT→letermovir

WHAT IT ENCODES

Algorithm: stable disease → valganciclovir PO; severe/GI → IV ganciclovir; HCT prophylaxis → letermovir; refractory CMV ≥2 weeks → genotypic resistance testing (UL97), then foscarnet/maribavir.

BOARD TRAP

The single most-tested branch: refractory viremia after 2 weeks of adequate therapy = resistance testing, not just escalate dose.